Ensemble Methods

TP – Risk certificates for ensemble methods

M2 Computer Science – MALIA

Guillaume Metzler
Institut de Communication (ICOM)
Université de Lyon, Université Lumière Lyon 2
Laboratoire ERIC UR 3083, Lyon, France

guillaume.metzler@univ-lyon2.fr

Abstract. In this practical session you will compute PAC-Bayesian risk certificates for random forests, Gibbs posteriors, boosted ensembles and linear classifiers on the datasets of the course, learn the weights of majority votes by minimizing these certificates, use them for model selection, and compare them with the classical hold-out certificate. The session relies on notebooks 1 to 5 and on the module `pbtools.py`. The work is organized in **four sessions of three hours**, each starting on a new page with its goals, the sections of the lecture notes to reread, a suggested timing and the results to hand in; parts are numbered by session (Part 2.3 is the third part of Session 2). The sessions build on each other, but the parts inside a session are independent after its first part and can be split among the members of a group. Parts marked (⋆) are longer.

Session	Theme	Parts
1	Certifying a classifier and a forest	warm-up on the kl, selection bias, OOB certificates, trees and bootstrap size
2	Self-bounding learning with forests	learning the weights, exact kl minimization, strength vs diversity, model selection
3	Gibbs posteriors and hard cases	temperature of a Gibbs posterior, hold-out vs self-bounding, imbalance, boosting
4	Linear classifiers and synthesis	PBGD and Monte Carlo certificates, data-dependent priors, synthesis, report

Session 1 – Certifying one classifier and a random forest

Goal. By the end of the session you should be able to compute a test-set bound by hand and with the kl inversion, understand selection bias, and compute the first-order, tandem and C-bound certificates of a random forest from its out-of-bag samples.
To reread before the session. Sections 2, 5 and 7.2 of the notes; notebooks 1 and 3.
Timing (3 hours).

Part 1.1	20 min
Part 1.2	50 min
Part 1.3	60 min
Part 1.4	50 min

To hand in at the end of the session. Your notebook, with the table of Part 1.3 on your 10–15 datasets and the table of Part 1.4.

Part 1.1 – Setting up

- Download the datasets (`datasets.zip`) and the loading script `prepdata.py` of the first practical session, and the module `pbtools.py`. The function `data_recovery(name)` returns a pair `(X, y)` with labels in $\{0, 1\}$: convert them to $\{-1, +1\}$ and standardize the features.
- Read `pbtools.py`: every function is short. It provides the binary kl and its inversions (`kl_bin`, `kl_inv_upper`, `kl_inv_lower`), the classical bounds (`bound_seeger`, `bound_mcallester`, `bound_catoni`, `bound_lambda`), a random forest keeping track of its out-of-bag samples (`BaggedTrees` with `oob_losses`, `oob_tandem`, `oob_disagreement`), the majority-vote certificates (`fo_bound`, `tnd_bound`, `cbound_pb`), two optimizers (`optimize_fo`, `optimize_tnd`) and the gradient of the kl-form certificates (`certificate_and_grad`).
- Use **at least 10 datasets**, balanced and imbalanced, for instance `australian`, `balance`, `bupa`, `german`, `heart`, `iono`, `pima`, `sonar`, `wdbc`, `spambase` (balanced group) and `abalone8`, `pageblocks`, `satimage`, `segmentation`, `yeast3` (imbalanced group).

> **Good practice.** Fix all the random seeds, and write one function per experiment that returns a table: you will rerun them many times. Before computing a certificate, write down on paper *which examples* each voter has seen: this single question avoids most of the mistakes of this session.

Part 1.2 – Warm-up: inverting the kl and certifying one classifier (Section 2)

1. Implement the upper inversion $\overline{\mathrm{kl}}^{-1}(\hat{q}, \varepsilon)$ by bisection (Algorithm 1 of the notes) and check it against `kl_inv_upper`. How many iterations are needed for a precision of 10^{-9}?
2. For $m \in \{100, 1000\}$, $\delta = 0.05$, $\varepsilon = \ln(1/\delta)/m$ and $\hat{q} \in \{0, 0.01, 0.05, 0.1, 0.3, 0.5\}$, compare three upper confidence bounds on the risk of a *fixed* classifier: Hoeffding $\hat{q} + \sqrt{\varepsilon/2}$, the kl inversion, and the refined Pinsker relaxation $\hat{q} + \sqrt{2\hat{q}\varepsilon} + 2\varepsilon$. When does the kl make a difference?
3. On `australian`, split the data in two halves. Learn a decision tree of depth 3 on the first half and compute the Hoeffding and kl test-set bounds of its risk on the second half.
4. Now learn trees of depths $1, \ldots, 10$ on the first half, keep the one with the smallest error on the second half, and compute its kl bound as if it had been fixed in advance. Why is this "certificate" not valid? Correct it.

Part 1.3 – Certificates of a standard random forest (Sections 5 and 7.2)

For each dataset, repeat 5 times: split the data into 70% training / 30% test (stratified) and learn a `BaggedTrees` forest of 100 trees on the training set with bootstrap samples of size $m/2$.

1. Compute the test risk of the uniform majority vote, the test Gibbs risk and disagreement, and the three OOB certificates (first-order, tandem, C-bound) of the uniform posterior, with $\delta = 0.05$. Report also the sizes n_1 (smallest OOB set) and n_2 (smallest intersection of two OOB sets).
2. Check that the certificates are always above the test risk. Which bound is the tightest, on which datasets? Relate your answer to the disagreement of the trees (Proposition 5.7) and to n_1, n_2.
3. Compare the test risk of `BaggedTrees` with `sklearn.ensemble.RandomForestClassifier` (full bootstrap). How much accuracy do we lose by using bootstrap samples of size $m/2$?

Part 1.4 – Number of trees and size of the bootstrap samples (Section 7.2)

On `spambase` (3 splits), learn forests with $n \in \{10, 50, 100, 200\}$ trees and bootstrap samples of size $k = f\,m$ with $f \in \{0.3, 0.5, 0.8, 1\}$.

1. Before running anything, predict how n_1 and n_2 depend on f (compute $(1 - 1/m)^{fm}$ and its square), and whether they depend on n.
2. Report the test risk of the uniform vote, n_1, n_2 and the FO and TND certificates. Which configuration gives the best vote? the best certificate? Is there a configuration that is good for both?

Session 2 – Self-bounding learning with random forests

Goal. By the end of the session you should be able to learn the weights of a forest by minimizing a certificate, compare relaxed and exact objectives, relate strength and diversity to the certificates, and select a model with certificates instead of cross-validation.
To reread before the session. Sections 5, 6 and 7.2 of the notes; notebook 4.
Timing (3 hours). Part 2.1 60 min
Part 2.2 45 min
Part 2.3 30 min
Part 2.4 45 min
To hand in at the end of the session. Your notebook, with the table of Part 2.1 with means and standard deviations, and the model-selection table of Part 2.4.

Part 2.1 – Learning the weights (Sections 6 and 7.2)

1. On each dataset, compute the posteriors minimizing the first-order bound (`optimize_fo`) and the tandem bound (`optimize_tnd`). Report the test risk of the corresponding votes, their certificates and the *effective number of trees* $1/\sum_t \rho_t^2$.
2. Produce a table with one row per dataset and, for each posterior (uniform, FO, TND), the mean and standard deviation over the 5 splits of the test risk and of the certificate. For the uniform posterior, report the best of the three certificates: what correction does this choice require?
3. Which posterior gives the best vote? the best certificate? Are the conclusions of notebook 4 confirmed?

Part 2.2 – Minimizing the kl certificate directly (Section 6.6)

The optimizers of Part 2.1 minimize a *relaxation* of the certificate (PAC-Bayes-λ). Since the certificate is valid for any posterior, we can also minimize the kl-form certificate itself.

1. Using `certificate_and_grad` (value and gradient of $4\,\overline{\mathrm{kl}}^{-1}(\rho^\top \widehat{\mathbf{L}}\rho, \varepsilon(\rho))$ with respect to the softmax logits, Proposition 6.6), implement Adam (300 iterations, step 0.05, starting from the uniform posterior).
2. On `spambase`, `satimage`, `wdbc` and `pageblocks`, compare the uniform posterior, the TND-optimized posterior of Part 2.1 and the posterior found by Adam: certificate, test risk, effective number of trees. Plot the certificate along the iterations.
3. Does the choice of the optimizer, of the step size or of the number of iterations require a union bound? Why?

Part 2.3 – Strength versus diversity (Section 5)

1. On `spambase` and `german`, learn forests with `max_features` $\in \{1, \sqrt{d}, d/2, d\}$ (3 splits). Report the test Gibbs risk, the test disagreement, the test risk of the vote and the OOB tandem and C-bound certificates.
2. Describe how strength and diversity evolve. Which value gives the best vote? Do the certificates follow the risk of the vote?

Part 2.4 – Self-bounding model selection (Section 6)

1. On `spambase`, `pima`, `wdbc` and `german`, learn $K = 5$ forests with `max_depth` $\in \{1, 2, 4, 8, \mathrm{None}\}$, minimize the tandem bound of each of them and compute its certificate with δ/K. Select the depth with the smallest certificate.

2. Compare with the depth selected by 5-fold cross-validation of the uniform forest on the training set, and with the best depth in hindsight (smallest test risk). Report the test risks of the selected models.
3. Why must the certificates be computed with δ/K? When does this selection rule fail, and what should be done then?

Session 3 – Gibbs posteriors, hold-out, imbalance and boosting

Goal. By the end of the session you should be able to build voters so that the prior is legitimate, compare self-bounding and hold-out certificates, certify the balanced error, and understand why boosted votes are hard to certify.
To reread before the session. Sections 4, 6 and 7.3 of the notes; notebooks 2 and 4.
Timing (3 hours).

	Part 3.1	50 min
	Part 3.2	35 min
	Part 3.3	50 min
	Part 3.4	45 min

To hand in at the end of the session. Your notebook, with the figure of Part 3.1 and the tables of Parts 3.2 to 3.4.

Part 3.1 – The Gibbs posterior and its temperature (Section 4)

We now build the voters ourselves, so that the prior can be fixed before seeing the data used in the bound.

1. On `wdbc` and `australian`, keep 20% of the data to build the voters: all the decision stumps $x \mapsto s\ \mathrm{sign}(x_j - \theta)$, $s \in \{\pm 1\}$, whose thresholds θ are the quantiles $10, 25, 50, 75, 90\%$ of feature j on this part. Split the remaining data into a sample S (60%) used by the bound and a test sample.

2. With the uniform prior, compute the Gibbs posterior $Q_\lambda \propto e^{-\lambda m \widehat{R}_S(h)}$ for a grid of 60 values of λ between 10^{-3} and $10^{1.5}$, and plot its empirical Gibbs risk, $\mathrm{KL}(Q_\lambda \| P)/m$ and Seeger's bound as functions of λ.

3. Report the best certificate, the corresponding test Gibbs risk and test risk of the vote, and compare with the Occam bound of the stump with the smallest empirical risk. Why is choosing λ on the grid free?

> **Numerical tip.** Compute the Gibbs posterior with a stable softmax, $Q_\lambda = \mathrm{softmax}(\ln P - \lambda m \widehat{R}_S)$, which subtracts the maximum before exponentiating; the naive formula underflows as soon as λm is large.

Part 3.2 – Hold-out certificate versus self-bounding certificate (Section 6.4)

1. *Self-bounding:* learn a forest on the whole training set, minimize its tandem bound, and report its TND certificate. *Hold-out:* learn the same forest on 70% of the training set and compute the kl test-set bound of its (uniform) vote on the remaining 30%. Compare the test risks and the certificates on `spambase`, `satimage`, `pageblocks`, `abalone8`, `wdbc`, `german`.

2. Which approach gives the better vote? the better certificate? Explain the gap between the two certificates.

3. In which situation would you nevertheless recommend the self-bounding approach?

Part 3.3 – Imbalanced datasets and the balanced error (Section 6)

The bounds of this course control the 0-1 risk. On imbalanced datasets a trivial classifier predicting the majority class has a small 0-1 risk.

1. On the imbalanced datasets, compare the certificates of the uniform forest with the risk of the trivial classifier. Are the certificates informative?

2. Obtain a certificate on the *balanced* error $\frac{1}{2}(\mathbb{P}(h(x) \neq y \,|\, y = +1) + \mathbb{P}(h(x) \neq y \,|\, y = -1))$: apply the OOB bounds separately to the training examples of each class (with n_+ and n_- OOB examples) and use a union bound. Implement it and comment.

Part 3.4 – Boosting (Section 7.3)

AdaBoost also outputs a weighted majority vote. To obtain a certificate, the voters must be evaluated on examples independent of them.

1. Split the training set into two halves S_1, S_2. Run `AdaBoostClassifier` with decision stumps (100 rounds) on S_1, and consider the learned stumps as the voters, with the uniform prior.
2. Compute on S_2 the FO and TND certificates of the posterior $\rho_t \propto \alpha_t$ given by AdaBoost (with $\delta/2$ each), and of the posteriors minimizing the two bounds. Compare the test risks with those of AdaBoost trained on the whole training set.
3. Why are the certificates of the AdaBoost posterior loose, although AdaBoost is accurate? *Hint:* compute the Gibbs risk of the stumps on S_2 and the KL of the AdaBoost posterior.

Session 4 – Linear classifiers, data-dependent priors and synthesis

Goal. By the end of the session you should be able to implement PBGD, certify a Gaussian posterior exactly and by Monte Carlo, learn a prior on part of the data, and put all the results of the four sessions together.
To reread before the session. Sections 4.2, 4.3, 6.4 and 7.4 of the notes; notebook 5.
Timing (3 hours). Part 4.1 75 min
Part 4.2 50 min
Part 4.3 30 min
Part 4.4 25 min
To hand in at the end of the session. Your notebook, with the tables of Parts 4.1 and 4.2, the decision table of Part 4.3, and the outline of the final report.

Part 4.1 – (⋆) PAC-Bayesian learning of linear classifiers and Monte Carlo certificates (Sections 4.3 and 7.4)

Consider linear classifiers $h_{\mathbf{v}}(x) = \operatorname{sign}\langle \mathbf{v}, \tilde{x}\rangle$ on the standardized features augmented with a constant 1 ($\tilde{x} = (x, 1)$), the prior $P = \mathcal{N}(\mathbf{0}, I)$ and the posteriors $Q_{\mathbf{w}} = \mathcal{N}(\mathbf{w}, I)$.

1. Implement the exact empirical Gibbs risk $\widehat{R}_S(G_{Q_{\mathbf{w}}}) = \frac{1}{m}\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle / \|\tilde{x}_i\|)$ and the PBGD objective $C\sum_i \Phi(-y_i\langle \mathbf{w}, \tilde{x}_i\rangle / \|\tilde{x}_i\|) + \frac{1}{2}\|\mathbf{w}\|^2$ with its gradient; minimize it with L-BFGS, starting from a linear SVM.

2. On `australian`, `pima`, `wdbc`, `spambase`, compute Seeger's certificate on the Gibbs risk for 13 values of C in $[10^{-2}, 10^2]$, select the best one, and compare the test risk of the vote $\operatorname{sign}\langle \mathbf{w}, \tilde{x}\rangle$ with a linear SVM ($C = 1$). Plot the certificate as a function of C.

3. Pretend that the Gibbs risk has no closed form: estimate it with $N = 2000$ classifiers drawn from $Q_{\mathbf{w}}$ and compute the Monte Carlo certificate (Section 4.3), with $\delta/2$ for each step. How much is lost?

4. Why can C be selected by minimizing the certificate without any union bound?

Part 4.2 – (⋆) Data-dependent priors for linear classifiers (Sections 4.2 and 6.4)

1. Split the training set into S_1 (a fraction f) and S_2. Learn a prior centre $\mathbf{w}_0$ on S_1 (PBGD with $C = 1$), then run PBGD on S_2 with the prior $\mathcal{N}(\mathbf{w}_0, I)$, i.e. with the regularizer $\frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$, and compute Seeger's certificate on S_2 only ($\mathrm{KL} = \frac{1}{2}\|\mathbf{w} - \mathbf{w}_0\|^2$), selecting C on the grid of Part 4.1.

2. Compare $f \in \{0, 0.2, 0.5\}$ ($f = 0$ is Part 4.1) on the four datasets: size of S_2, KL, certificate, test risk.

3. Which examples may the posterior use? Why is the certificate valid although $\mathbf{w}_0$ depends on the data?

Part 4.3 – Synthesis: which certificate for which situation?

Using your results of the four sessions (and not new computations), fill in the following decision table. For each situation, say which guarantee you would report (test-set bound, first-order, tandem or C-bound certificate, Seeger's bound on a Gibbs classifier, with or without data-dependent prior), and justify your choice in one sentence with a number from your experiments.

situation	guarantee to report, and why
a forest has been learned, and 30% of the data can be set aside	
a small dataset (a few hundred examples)	
the weights of the vote are learned from the data	
several models are compared and one is selected	
an imbalanced problem	
a boosted vote of stumps	
a stochastic (Gibbs) linear classifier	

Part 4.4 – Report (to finish after the session)

Summarize your results in a short report (3–4 pages) containing the tables of Parts 1.3 and 2.1, two figures of your choice and a discussion. As in the first practical session, compare methods that are comparable and give standard deviations; remember that a certificate is not an estimate of the risk, and comment on its tightness, not only on its value; finally, report every choice made using the data (grids, splits, selected bounds), since each of them may require a union bound.